

Derivative Placement in Distributed Nonlinear Finite Elements: Conditional Equivalence and Distributed Verification

Michal Bérés

Institute of Geonics of the Czech Academy of Sciences, Ostrava, Czech Republic

Department of Applied Mathematics, Faculty of Electrical Engineering and Computer Science,

VSB – Technical University of Ostrava, Ostrava, Czech Republic

`michal.beres@ugn.cas.cz`

Abstract

Second derivatives of nonlinear finite-element energies may be constructed at the element, constitutive, or assembled sparse-operator level. Element and constitutive differentiation are mathematically equivalent when they differentiate the same quadrature functional, affine constraint map, and smooth constitutive branch. Colored recovery yields the same sparse operator under the additional requirements of a complete structural pattern, a valid column- interference coloring, consistent overlap values, and unique global row ownership. We state these conditional results and implement the three routes with JAX and PETSc.

The numerical study separates derivative correctness from nonlinear and performance claims. Across 15 prescribed P_1 , P_2 , and P_4 states, the largest element-versus-constitutive residual and Hessian discrepancies are 2.35×10^{-16} and 2.34×10^{-16} . For distributed colored recovery, prescribed tangent actions agree on elastic and mixed states over one, two, and four Message Passing Interface (MPI) ranks; the largest cross-rank discrepancy is 2.64×10^{-16} , and direct compressed sparse row (CSR) comparisons pass at one rank. Material-point tests cover all five branches of a three-dimensional Mohr–Coulomb endpoint surrogate while excluding ordinary Hessian claims at switches. Independent scalar and hyperelastic manufactured problems recover the expected linear-element spatial behavior.

Fixed-state quadrature comparisons show that small potential differences can coexist with appreciable residual and Hessian-action differences; for degree-four elements on the base mesh, the latter reach 2.68% and 6.91%, respectively. The results therefore support the conditional derivative identities and distributed construction on the tested states. We make no universal route-ranking, crossover, timing, or scaling claim.

1 Introduction

Nonlinear finite-element models are often defined by compact energy or constitutive programs, while their solution requires distributed sparse matrices, Krylov methods, and nonlinear globalization. Automatic differentiation (AD) removes many manual derivative formulas, but it does not remove the choice of where differentiation occurs. A tangent may be obtained from the complete element energy, from a low-dimensional constitutive density, or from colored actions of the assembled Hessian. Each route exposes a different combination of local arithmetic, memory traffic, sparse insertion, and communication.

This choice is scientific only if the compared routes represent the same discrete problem. Quadrature, essential constraints, affine lifts, active constitutive branches, engineering-shear conventions, and state ordering must remain fixed. Distributed colored recovery introduces further requirements: the structural pattern must be complete, same-color columns must not interfere in an owned row, and every ghost value needed by a local action must be available. Agreement of endpoint energies alone does not verify these conditions.

Existing frameworks establish the principal component technologies. High-level finite-element systems automate variational forms and parallel assembly [1, 2]; JAX-based methods provide differentiable mechanics and inverse problems [3, 4]; and bridge architectures combine differentiable local programs with established sparse solvers [5–7]. Recent work also develops locality-aware element differentiation, global AD with sparse tangent materialization, and automatic sparsity detection [8–10]. Consequently, the novelty of the present study is not JAX–PETSc coupling, constitutive AD, or graph coloring in isolation.

We instead ask two narrower questions. First, which assumptions guarantee that element AD, constitutive AD, and colored sparse recovery differentiate one quadrature-defined potential? Second, which fixed-state and distributed tests expose errors in branch selection, local contraction, sparse recovery, or Message Passing Interface (MPI) ownership? Both questions admit mathematical statements and finite correctness tests. Performance selection is outside the scope of the present study.

1.1 Contributions

The paper makes four contributions.

1. We formulate a branchwise equivalence result for element and constitutive differentiation of a fixed quadrature potential. The statement includes the constrained affine map and excludes branch switches and unresolved repeated spectra.
2. We state the interference, overlap, and row-ownership conditions under which colored Hessian actions recover the same distributed sparse operator.
3. We provide an evidence hierarchy that separates analytic and manufactured verification, canonical fixed-state MPI checks, and bounded nonlinear-outcome studies. Each numerical statement is tied to the algebraic object and admission conditions actually tested.
4. We exercise this hierarchy on smooth scalar energies, orientation-preserving hyperelasticity, and a synthetic branch-structured Mohr–Coulomb endpoint potential. The latter is used to test derivative placement and is not presented as incremental plasticity validation.

The resulting evidence is deliberately asymmetric. The manufactured and analytic verifiers pass for the tested smooth problems. Element and constitutive differentiation agree on 15 prescribed finite-element states, and distributed colored recovery matches the prescribed element-route tangent actions for $P_1(L_1)$ and $P_2(L_1)$ on one, two, and four ranks. Direct complete sparse-matrix comparisons additionally pass at one rank. A separate checksum-verified workstation study corroborates fixed-state tangent-action agreement. We report no route timing and claim neither a fastest route nor a crossover or scaling law.

Section 2 positions this scope against the closest software and methodological work. Sections 3 and 4 state the mathematical and distributed contracts. Section 5 defines the retained problems, and Section 6 presents independent verification designs and results. The remaining numerical outcomes and their limitations appear in Section 7; Sections 8 and 9 summarize the resulting claim boundary.

2 Related Work

The closest literature separates into finite-element automation, differentiable mechanics, constitutive differentiation, and sparse derivative recovery. We organize it by the location of differentiation because software feature inventories do not determine whether two routes represent the same discrete functional.

The FEniCS project established symbolic weak-form specification and generated finite-element kernels [1]. DOLFINx extends this model with data-oriented parallel interfaces [2]. Automated adjoint systems operate at a higher program level: `dolfin-adjoint` and `pyadjoint` derive sensitivities of finite-element programs [11, 12], while `cashocs` provides adjoint-based optimal-control and shape-optimization workflows [13]. These systems motivate automation, but the present work concerns primal residual and tangent construction inside a Newton solve.

JAX provides composable program transformations for differentiation, vectorization, and compilation [14, 15]. JAX-FEM applies them to forward and inverse finite-element mechanics [3], and second-order implicit differentiation extends this setting to Hessian-vector products for differentiable physics [4]. AutoPDEx and JAX-CPFEM illustrate related energy-based and constitutive applications [16, 17]. These results establish the viability of differentiable finite elements; they do not compare the three derivative locations studied here under one distributed algebraic contract.

Bridge architectures are particularly close. Firedrake–JAX differentiates partial differential equation solves through tangent and adjoint equations [5]. FEniCSx external operators admit general quadrature-point programs, including a JAX-differentiated Mohr–Coulomb model with Taylor tests [6]. JetSCI combines JAX-local discretizations with PETSc-owned distributed solves [7]. Thus neither constitutive AD nor the JAX–PETSc architecture is new here. Our distinction is the fixed-functional equivalence result and an evidence hierarchy for comparing element, constitutive, and colored routes.

Recent systems narrow this distinction further. Locality-aware AD performs per-element forward differentiation and sparse Hessian assembly on graphics processing units [8]. The `tatva` framework applies AD to a global energy and uses matrix-free products or coloring to materialize sparse tangents [9]. Hill and Dalle combine automatic local or global sparsity detection, matrix coloring, and sparse differentiation [10]. These papers establish both local and global AD strategies. The present contribution is therefore limited to explicit branchwise equivalence assumptions, distributed finite-element ownership conditions, and a fixed-state verification hierarchy on processor-based MPI systems.

Sparse recovery itself is classical. Curtis, Powell, and Reid introduced directional probing for sparse Jacobians [18]; Coleman and Moré related sparse Jacobian and Hessian estimation to graph coloring [19, 20]. PETSc supplies the distributed vectors, matrices, Krylov methods, and preconditioners used in our implementation [21, 22]. We do not claim these algorithms as new. We use them to expose the assumptions needed to turn local Hessian actions into a canonical distributed matrix and to separate finite correctness evidence from performance claims.

The branch-structured benchmark is motivated by Mohr–Coulomb slope stability, but its evidentiary role is intentionally narrower. Incremental plasticity, return mapping, and consistent linearization provide the continuum reference frame [23–25]. Variational strength reduction and recent three-dimensional continuation methods provide application context [26, 27]. Our implemented object is a synthetic endpoint potential with selected branches. It tests derivative placement; it does not validate a path-consistent constitutive history or a factor of safety.

3 Discrete Derivative Equivalence

We first define the algebraic object differentiated by every route. Let $V_{h,0} = \text{span}\{\varphi_i\}_{i=1}^n$ be the finite-element space satisfying homogeneous essential conditions, and let $x \in \mathbb{R}^n$ contain its free scalar coefficients. Essential data enter through an affine lift $\bar{u}_h(\theta)$, where θ collects fixed load and material parameters. The corresponding state is

$$u_h(x; \theta) = \bar{u}_h(\theta) + \sum_{i=1}^n x_i \varphi_i. \quad (1)$$

For element e , let R_e restrict the global free vector to the coefficients whose basis functions intersect e , and let L_e place these coefficients in the chosen local component ordering. We define

$$u_e(x; \theta) = L_e R_e x + b_e(\theta) =: A_e x + b_e(\theta), \quad A_e := L_e R_e, \quad (2)$$

where $b_e(\theta)$ contains the element coefficients of the affine lift. Thus A_e is fixed for a given constrained space and permits arbitrary permutations of the free degrees of freedom (DOFs) without assigning coefficients to constrained DOFs.

Let B_{eq} be the fixed linear map from the local coefficients to the complete quadrature-point argument required by the density. This argument may stack a field value and its gradient, as for Ginzburg–Landau, or reduce to a strain or deformation gradient in mechanics. We fix the quadrature rule, weights, local material data m_{eq} , and a deterministic first-match selector $\beta_{eq}(x; \theta)$ evaluated from this argument and the material data. We define the local quadrature jet

$$z_{eq}(x; \theta) := B_{eq}(A_e x + b_e(\theta)). \quad (3)$$

If $\mathcal{U}_\alpha(m)$ denotes the domain of constitutive branch α , we take the admissible free-space domain to be

$$\mathcal{D}_h(\theta) := \left\{ x \in \mathbb{R}^n \left| \begin{array}{l} \beta_{eq}(x; \theta) \text{ is defined, } z_{eq}(x; \theta) \in \mathcal{U}_{\beta_{eq}(x; \theta)}(m_{eq}), \\ \mathcal{W}_{\beta_{eq}(x; \theta)}(z_{eq}(x; \theta); m_{eq}) \in \mathbb{R} \text{ for every } (e, q) \end{array} \right. \right\}. \quad (4)$$

This definition includes model-specific restrictions, such as positive deformation Jacobians in hyperelasticity. The discrete element contribution is

$$\Phi_e(x; \theta) := \sum_{q=1}^{n_q} w_{eq} \mathcal{W}_{\beta_{eq}(x; \theta)}(z_{eq}(x; \theta); m_{eq}), \quad (5)$$

and the global free-space potential is

$$\Pi_h(x; \theta) := \sum_{e \in \mathcal{T}_h} \Phi_e(x; \theta) - f_h(\theta)^\top x. \quad (6)$$

Here the linear external functional $\ell_h(\cdot; \theta)$ is represented on the free basis by $(f_h(\theta))_i := \ell_h(\varphi_i; \theta)$. Its value on the affine lift is independent of x and may therefore be omitted together with any other additive term that depends only on prescribed data. The discrete optimization problem is

$$\min_{x \in \mathcal{D}_h(\theta)} \Pi_h(x; \theta). \quad (7)$$

On an open subset of $\mathcal{D}_h(\theta)$ where the selected branches are twice continuously differentiable, we define

$$g_h(x; \theta) := \nabla_x \Pi_h(x; \theta), \quad H_h(x; \theta) := \nabla_x^2 \Pi_h(x; \theta). \quad (8)$$

The algorithms seek admissible stationary points, characterized by $g_h(x; \theta) = 0$ at branch-interior states. A small residual alone neither certifies a local minimizer nor excludes another stationary point. Such a claim would additionally require an appropriate second-order or global argument, which we make only when it is verified for the case in question. The symbol $P_k(L_\ell)$ denotes degree- k Lagrange elements on mesh level L_ℓ .

3.1 Element and constitutive differentiation

Element AD applies first- and second-order differentiation directly to Φ_e . At a fixed state $x_\star$, write $\beta_{eq}^\star := \beta_{eq}(x_\star; \theta)$. Constitutive AD differentiates the same selected scalar density at the same quadrature jet. We define

$$\begin{aligned} z_{eq}^\star &:= z_{eq}(x_\star; \theta), \\ \sigma_{eq}^\star &:= \partial_z \mathcal{W}_{\beta_{eq}^\star}(z_{eq}^\star; m_{eq}), \\ C_{eq}^\star &:= \partial_{zz}^2 \mathcal{W}_{\beta_{eq}^\star}(z_{eq}^\star; m_{eq}). \end{aligned} \quad (9)$$

Here $\sigma_{eq}^\star$ denotes the vector conjugate to the complete local jet; it is stress-like when that jet is a strain or deformation gradient. The branch-stability assumption below makes $\beta_{eq}(x; \theta) = \beta_{eq}^\star$ on a neighborhood of $x_\star$; the selector is therefore locally constant during branch-interior differentiation.

Proposition 1 (Fixed-functional branchwise equivalence). *Fix $x_\star \in \mathcal{D}_h(\theta)$. Assume that, on a neighborhood of $x_\star$, (i) both routes use the same scalar densities, quadrature weights, material and history data; (ii) $u_e(x) = A_e x + b_e$ is affine; (iii) every B_{eq} is fixed; (iv) each selected density is twice continuously differentiable; (v) the branch labels and principal-value ordering do not change; and (vi) both routes use the same tensor and engineering-shear conventions. Then element AD and constitutive AD give identical element residuals and Hessians in exact arithmetic:*

$$\begin{aligned}\nabla_x \Phi_e(x_\star; \theta) &= \sum_q w_{eq} A_e^\top B_{eq}^\top \sigma_{eq}^\star, \\ \nabla_x^2 \Phi_e(x_\star; \theta) &= \sum_q w_{eq} A_e^\top B_{eq}^\top C_{eq}^\star B_{eq} A_e.\end{aligned}\tag{10}$$

Consequently, assembling either set of element objects gives the same global free-space gradient and Hessian of (6).

Proof. The affine map has zero second derivative. Applying the chain rule first to $z_{eq}(x; \theta)$ and then to every summand of (5) gives (10). Summation and the fixed free-space assembly map preserve the identities. $\square$

In particular, the linear external load contributes a constant gradient and no Hessian. The assembled derivatives at $x_\star$ are therefore

$$\begin{aligned}g_h(x_\star; \theta) &= \sum_{e \in \mathcal{T}_h} \sum_q w_{eq} A_e^\top B_{eq}^\top \sigma_{eq}^\star - f_h(\theta), \\ H_h(x_\star; \theta) &= \sum_{e \in \mathcal{T}_h} \sum_q w_{eq} A_e^\top B_{eq}^\top C_{eq}^\star B_{eq} A_e,\end{aligned}\tag{11}$$

because

$$\nabla_x [-f_h(\theta)^\top x] = -f_h(\theta), \quad \nabla_x^2 [-f_h(\theta)^\top x] = 0.\tag{12}$$

Proposition 1 is conditional. It does not apply at a branch switch, under an unresolved principal-value ordering, or when the compared codes change quadrature or constraints. It also establishes the derivatives of the implemented scalar potential, not equivalence to an incremental return-mapping model.

3.2 Colored sparse recovery

Let $H := H_h(x; \theta)$ and let $\rho : \{1, \dots, n\} \rightarrow \{1, \dots, N_{\text{rank}}\}$ assign every global free row to one mathematical owner. We set $I_r := \rho^{-1}(r)$ and let $\mathcal{P}_r$ contain the structural entries (i, j) of those rows. The local element set $\mathcal{E}_r$ contains every element incident on I_r , and J_r contains every free column appearing in those elements. We write $H^{(r)}$ for the Hessian of the sum over $\mathcal{E}_r$, restricted to J_r . Because all element contributions incident on an owned row are present, $H_{ij}^{(r)} = H_{ij}$ for $(i, j) \in \mathcal{P}_r$.

The rank-local column-interference graph joins columns $j, k \in J_r$ whenever some owned row $i \in I_r$ satisfies $(i, j), (i, k) \in \mathcal{P}_r$. A local color class $\mathcal{C}_{r,c}$ defines the overlap seed

$$v_{r,c} := \sum_{j \in \mathcal{C}_{r,c}} e_j, \quad y_{r,c} := H^{(r)} v_{r,c}.\tag{13}$$

This construction is the direct Hessian analogue of classical sparse derivative probing [18, 20].

Proposition 2 (Owned-row colored recovery). *For every rank r , assume that (i) $\mathcal{P}_r$ contains every active nonzero (i, j) of each owned row; (ii) $\mathcal{E}_r$ contains every element contributing to those rows, with complete state, affine-lift, material, and selected-branch data on its overlap; (iii) J_r contains every column in $\mathcal{P}_r$, and each owned row contains at most one structural entry from any $\mathcal{C}_{r,c}$; and (iv) $H^{(r)} v_{r,c}$ is evaluated exactly with the complete overlap seed. Then, only for $(i, j) \in \mathcal{P}_r$ with $j \in \mathcal{C}_{r,c}$, assigning $H_{ij} := (y_{r,c})_i$ recovers every owned structural entry exactly. The union of the owned rows is the canonical distributed matrix H .*

Proof. For an owned row i , the identity $(y_{r,c})_i = \sum_{j \in \mathcal{C}_{r,c}} H_{ij}^{(r)}$ contains at most one structural nonzero by assumption (iii). Assumptions (i), (ii), and (iv) make the local sum complete and give $H_{ij}^{(r)} = H_{ij}$ on the owned pattern. The restricted assignment therefore recovers that entry, and unique ownership assembles each recovered row once. $\square$

In the implementation, the nonoverlapping PETSc row ranges realize the map ρ . Proposition 2 depends only on the unique mathematical assignment, not on this particular library convention.

Proposition 3 (Component-lifted scalar coloring). *Let $S \in \{0, 1\}^{N_s \times N_s}$ be the scalar support pattern, with $S_{ab} = 1$ whenever scalar basis functions a and b occur together on at least one element. Suppose that the structural pattern of a d -component vector Hessian is a subpattern of $S \otimes \mathbf{1}_{d \times d}$. If c_s is a column-intersection coloring of S , then*

$$c(a, k) := d c_s(a) + k, \quad k \in \{0, \dots, d-1\},\tag{14}$$

is a valid column-intersection coloring of every constrained vector subpattern.

Proof. Suppose that vector columns (a, k) and (b, ℓ) meet one structural row. If $a \neq b$, then scalar columns a and b meet the corresponding scalar support row, so $c_s(a) \neq c_s(b)$. If $a = b$, distinct vector columns have $k \neq \ell$. Equation (14) therefore assigns different colors in both cases. Removing constrained columns cannot introduce a conflict. $\square$

The proposition permits different rank-local color labels and counts because each rank evaluates its own complete incident-element Hessian and inserts only its owned pattern. It does not permit a missing incident element or structural edge, incomplete state or seed overlap, inconsistent branch data, or two same-color columns that meet an owned row. The implemented recovery uses AD Hessian-vector products; a finite-difference gradient probe would add its truncation error.

3.3 Mesh-aware nonlinear accuracy

Coefficient norms change with mesh density and therefore do not define a portable stopping contract. Let M_h be a symmetric positive-definite Riesz operator on the constrained free space. We use the primal and dual norms

$$\|s\|_{M_h} := (s^\top M_h s)^{1/2}, \quad \|g\|_{M_h^{-1}} := (g^\top M_h^{-1} g)^{1/2}. \quad (15)$$

The scalar problems use a row-sum lumped P_1 mass operator. Hyperelasticity and the branch-structured mechanics problem use the tangent of a reference elastic energy on the identical constrained free space. Before a reference tangent is admitted, a scale-aware symmetry check and a recorded direct factorization report zero negative and zero numerical zero pivots under the stated solver options. This is a numerical inertia check, not an analytic positive-definiteness proof. Each inverse-Riesz solve also passes an independently recomputed true-residual gate.

For nonlinear iteration $k \geq 0$, let $g_k := g_h(x_k; \theta)$ and define

$$R_k := \|g_k\|_{M_h^{-1}}, \quad U_k := \|x_k\|_{M_h}. \quad (16)$$

For $k \geq 1$, we additionally define

$$C_k := \|x_k - x_{k-1}\|_{M_h}, \quad D_k := |\Pi_h(x_k; \theta) - \Pi_h(x_{k-1}; \theta)|. \quad (17)$$

We report the dimensionless quantities

$$r_k^{\text{rel}} := \frac{R_k}{R_0}, \quad c_k^{\text{rel}} := \frac{C_k}{\max(u_{\text{ref}}, U_k)}, \quad (18)$$

where r_k^{rel} is defined only when $R_0 > 0$. The scale $u_{\text{ref}} > 0$ has the same units as the primal norm. It may equal U_0 when $U_0 > 0$; otherwise, an explicit positive scale must be prescribed independently.

The branch-structured nonlinear stopping rows use a gradient-only, initial-relative criterion with a prescribed target $\tau_R^{\text{rel}} > 0$. The terminal Boolean is

$$T_k := \{R_0 > 0\} \wedge \{r_k^{\text{rel}} \leq \tau_R^{\text{rel}}\}. \quad (19)$$

The absolute residual tolerance is zero, so no absolute fallback modifies (19). An iterate may terminate only after it is accepted and all required terminal data are finite, but these run-validity conditions are not alternative convergence measures. If $R_0 = 0$, the ratio is undefined and the solve lies outside this relative-only contract; such an initially stationary case must be documented separately.

The quantities D_k , C_k , U_k , and c_k^{rel} are reported as diagnostics; neither a correction threshold nor an energy-change threshold enters (19). The endpoint analysis independently recomputes R_k and verifies the selected branch, Riesz solve, state identity, and provenance; these admission checks do not alter the gradient-only stopping rule.

For hyperelasticity, the coefficient vector represents the free part of the deformation map under one fixed affine lift. Its reference-energy norm is used only within that representation and is not invariant under a change of lift. The endpoint record contains the selected Riesz operator, scale, free-space identity, numerical inertia record, inverse-solve diagnostics, and independently recomputed terminal norms.

The Riesz operators and tolerances are case-specific. The resulting residuals permit comparisons within the stated free-space representation; they do not establish a universal mesh-independent stopping threshold.

3.4 Evidence contract

Correctness precedes time. A fixed-state route row first passes the exact state identity, the common element-AD residual, four prescribed tangent actions, branch, and structural-covariate checks. The common residual verifies the state and shared assembly path; it is not an independent constitutive-gradient test. At the feasible one-rank P_1 points, the row also passes a direct complete compressed sparse row (CSR) comparison. Energy-change and correction quantities remain diagnostics and are not route-admission gates.

This hierarchy separates a proved identity under stated assumptions from verification on finite tested states. Evidence for either statement does not imply a performance ordering. In particular, near-equal energy does not replace a residual or tangent comparison, and one tangent probe does not establish complete matrix equality.

4 Distributed Construction and Numerical Contract

The implementation assigns local differentiable kernels to JAX and distributed algebraic objects to PETSc. Each MPI rank stores one contiguous interval of the free coefficient vector and the ghost values required by its overlap elements. The same ownership map is used for the residual, tangent rows, Riesz operator, and canonical state export. This common map is essential: comparing local arrays before undoing a route-specific permutation would not compare the same algebraic object.

4.1 Local derivative routes

The element route evaluates a scalar Φ_e and obtains its gradient and Hessian by JAX transformations. The constitutive route evaluates the same selected quadrature density, obtains σ_{eq} and C_{eq} from (9), and contracts them according to (10). For the endpoint-surrogate experiments, the residual is also evaluated by element AD; only the tangent differentiation boundary changes. This choice prevents a residual-formula difference from being hidden inside a tangent comparison.

The colored route first builds the complete structural pattern of every owned row. For the three-dimensional vector problem, it colors the active scalar-node support at distance two with deterministic lexical weights and lifts a scalar color $c_s(a)$ to displacement component k as $3c_s(a) + k$. The lift is validated against every scalar support row before differentiation. It is a conservative coloring of the vector column-interference graph by Proposition 3, including under componentwise constraints. Exact JAX Hessian-vector products are then evaluated in fixed batches of four colors, and each batch is inserted into the uniquely identified owned entries before the next batch is formed. The final incomplete batch is zero padded, so compilation shape does not depend on the color count. The implementation does not assume that different ranks obtain the same color labels. It relies instead on Proposition 2 and unique PETSc row ownership.

All three routes use binary64 arithmetic. Compilation and kernel warmup occur before measured repetitions. Compiled element, constitutive, and action kernels remain separate because their traces and intermediate arrays are different scientific costs. At the feasible one-rank P_1 route points, full-mesh compressed sparse row (CSR) arrays are written to transient disk-backed storage and compared in bounded chunks; the complete matrix norms and maximum entrywise defects are evaluated rather than sampled. Other full-mesh route points use four prescribed tangent actions, while the fixed-element verifiers compare complete local Hessians.

4.2 Rank-local mesh and sparse assembly

Let I_r denote rank r 's owned free DOFs and G_r its required ghosts. The retained overlap is

$$\mathcal{T}_r^{\text{loc}} := \{e : \text{dofs}(e) \cap I_r \neq \emptyset\}. \quad (20)$$

Before each local evaluation, a point-to-point exchange populates G_r and the affine lift supplies constrained values. Residual and tangent contributions are accumulated only for rows in I_r ; off-rank columns retain global indices. PETSc's distributed coordinate format is preallocated once and reused.

Scalar energy requires a separate ownership rule because overlap elements occur on more than one rank. Each element has a deterministic owner $o(e)$ and weight $\omega_{r,e} = 1$ if $o(e) = r$. Hence

$$\sum_r \sum_{e \in \mathcal{T}_r^{\text{loc}}} \omega_{r,e} \Phi_e = \sum_{e \in \mathcal{T}_h} \Phi_e. \quad (21)$$

This identity, the affine lift, the canonical free-DOF order, and the CSR indices are included in the distribution-equivalence test.

The distributed hyperelastic element path builds the structured mesh and overlap data rank locally. Its one-rank replicated construction is retained only as an algebraic comparator. The endpoint-surrogate meshes and material data are immutable inputs; quadrature rules have stable names and are recorded in every result.

4.3 Nonlinear and linear solvers

The nonlinear driver accepts either line-search Newton or a trust-region policy [28, 29]. Armijo backtracking is used for the controlled Newton comparisons. The reduced-subspace trust method minimizes the quadratic model on the span of the Newton and gradient directions and applies the same configured Armijo rule when a post-step search is requested. Thus a method label fixes the actual globalization algorithm rather than selecting an undocumented fallback.

Each accepted Newton direction is produced by a PETSc Krylov solve on the current tangent. The result record contains the requested and effective Krylov-solver tolerances, iteration count, convergence reason, and an independently recomputed true residual. A nonconverged Krylov solve, nonfinite direction, rejected line search, trust-region rejection cap, or nonlinear iteration cap remains a failure. Such a row is never converted into a successful observation with a large time.

The primary verification configurations use ordinary Krylov methods with hypre preconditioning or a sparse direct factorization. No preconditioner ranking is inferred in this paper. Reference-elastic Riesz operators are independent of the Newton tangent and remain fixed over the associated nonlinear trajectory.

4.4 Riesz operators and terminal recomputation

For a lumped scalar mass operator, the diagonal weights are assembled on the exact free space and must be finite and strictly positive. For mechanics, K_{ref} is copied from the exact discrete elastic tangent at the declared reference state. Hyperelasticity uses the identity deformation $y(X) = X$ with both end faces constrained. The branch-structured mechanics Riesz operator uses the elastic constitutive branch at zero displacement with the glued-bottom constraint.

The constrained matrix is first checked for scale-aware symmetry. A sparse direct factorization then reports its numerical inertia under the stored factorization and scaling options; admission requires

$$(n_-, n_0, n_+) = (0, 0, n_{\text{free}}). \quad (22)$$

The dual norm is evaluated by an independent conjugate-gradient (CG) solve with a recomputed true-residual threshold. PETSc options cannot silently modify this solve unless an explicit, namespaced override is recorded. At every terminal state, the driver re-evaluates the energy, gradient, primal state norm, dual residual, and correction from the final stored vector. A failed run therefore cannot inherit an initial or pre-step residual as its endpoint value.

4.5 Canonical records

Every runner writes strict JavaScript Object Notation (JSON): nonfinite sentinels are serialized as `null`, and the file becomes visible only after an atomic replacement. The schema records the experiment, case, route, repetition, MPI size, threads, input and code identities, solver controls, terminal status, norms, work counters, timing phases, and state hashes. Canonical NumPy `.npz` states use the original global ordering and include coordinate, connectivity, constraint, and metric metadata. State identities use the 256-bit Secure Hash Algorithm (SHA-256) applied to the array data type, shape, and raw binary64 bytes. Hash equality is an exact reproducibility check, not a numerical norm or an assertion of physical endpoint equivalence.

5 Verification Problems and Test Design

The numerical study uses four problem blocks. Smooth scalar manufactured solutions measure spatial consistency. Hyperelasticity supplies analytic stress/tangent checks, nonaffine convergence, and canonical distributed assembly tests. A synthetic Mohr–Coulomb endpoint potential exercises constitutive branches and high-order quadrature. Controlled Ginzburg–Landau runs probe an indefinite Hessian and globalization failure semantics.

5.1 Manufactured p -Laplace problem

Let $p = 3$ and $\Omega = (0, 1)^2$; the weak formulation follows the standard stationary p -Laplace setting [30, Ch. 2]. We consider

$$\mathcal{E}(u) := \int_{\Omega} \frac{1}{p} |\nabla u|^p - fu \, dx, \quad (23)$$

with exact Dirichlet data and source determined by

$$u_{\star}(x, y) := x + y + 0.1 \sin(\pi x) \sin(\pi y), \quad f := -\nabla \cdot (|\nabla u_{\star}| \nabla u_{\star}). \quad (24)$$

The gradient satisfies $|\nabla u_{\star}| \geq 1 - 0.1\pi > 0.685$, avoiding the degenerate zero-gradient Hessian during the consistency test. The independent verifier uses uniform P_1 triangles, a degree-five seven-point load/error rule, and a consistent NumPy/SciPy Newton tangent. It does not call the implemented JAX or PETSc assembly.

On the nonempty affine Dirichlet space, the gradient term is coercive and its strictly monotone derivative is injective on free variations; hence the finite-dimensional energy has a unique discrete minimizer. The measured quantities are the weighted L^2 error, the H^1 -seminorm error, the weak residual, the tangent symmetry defect, and the minimum element-gradient norm. The expected P_1 orders are two and one.

5.2 Manufactured Ginzburg–Landau problem

The real scalar energy with a source is

$$\mathcal{G}(u) := \int_{\Omega} \frac{\epsilon}{2} |\nabla u|^2 + \frac{1}{4} (u^2 - 1)^2 - fu \, dx, \quad \epsilon = 0.04. \quad (25)$$

We select

$$u_{\star}(x, y) := 0.8 + 0.1 \sin(\pi x) \sin(\pi y), \quad f := -\epsilon \Delta u_{\star} + u_{\star}(u_{\star}^2 - 1), \quad (26)$$

and impose its trace on $\partial\Omega$. The solve uses the same symmetric three-point triangle rule as the production JAX functional, while the error is evaluated by the seven-point rule. The exact solution satisfies $u_{\star} \geq 0.8 > 1/\sqrt{3}$, and the recorded computed nodal values also remain above this curvature threshold. Thus the tested states, rather than the exact solution alone, establish the positive-curvature restriction. This design measures spatial consistency without conflating it with selection among competing double-well basins.

The separate globalization study uses the zero-source functional implemented by the nonlinear driver and three checksum-identified initial states: a nominal state and two deterministic modal perturbations. It compares Newton–Armijo with the reduced-subspace trust method under the same generalized minimal residual (GMRES) method with hypre preconditioning, with five separate executions for each start–method pair. This design verifies bounded outcome semantics; it does not estimate a robustness probability.

5.3 Hyperelasticity

Each mechanics benchmark below uses an explicitly nondimensional computational system. Let $L_0 > 0$ and $S_0 > 0$ denote its reference length and stress, or equivalently energy-density, scales. We report X/L_0 , y/L_0 , W/S_0 , stress-like material parameters divided by S_0 , and body-force densities multiplied by L_0/S_0 . The stored arrays contain these scaled quantities, so the computational data take $L_0 = S_0 = 1$; we make no physical-unit or calibration claim.

Let $y : \Omega \rightarrow \mathbb{R}^3$ be the deformation map, $F := \nabla y$, and $J := \det F > 0$. We use the compressible neo-Hookean density

$$W(F) := C_1(\text{tr}(F^\top F) - 3 - 2 \log J) + D_1(J - 1)^2, \quad (27)$$

where $C_1 = 0.5$ and $D_1 = 5$ for the independent affine and manufactured verification problems on $\Omega = (0, 1)^3$. The distributed cantilever uses $\Omega = [0, 0.4] \times [-0.005, 0.005]^2$, $C_1 = 3.8462 \times 10^7$, and $D_1 = 8.3333 \times 10^7$ in the same scaled system; run records retain the full binary64 inputs. Its first Piola stress is $P := \partial W / \partial F$ [31, Chs. 4–6]. The density is smooth on $J > 0$ but nonconvex in the free coefficients.

The affine patch maps a unit cube by

$$y(X) = F_\star X + t, \quad F_\star = \begin{bmatrix} 1.15 & 0.08 & 0 \\ 0.02 & 0.92 & 0.05 \\ 0 & 0.03 & 1.08 \end{bmatrix}, \quad \det F_\star = 1.139187. \quad (28)$$

An independent analytic implementation assembles W , P , the material tangent, nodal internal forces, and constant boundary tractions. The JAX element energy is differentiated on the identical six-tetrahedron mesh. Translation modes and a superposed rigid rotation test objectivity and covariance.

The nonaffine manufactured deformation is

$$y_\star(X, Y, Z) := (X + 0.05 \sin(\pi X) \sin(\pi Y) \sin(\pi Z), Y, Z). \quad (29)$$

Its trace is imposed on the complete boundary, and the analytic body force is $b = -\text{Div } P(\nabla y_\star)$. The independent P_1 verifier uses uniform cube subdivisions 4, 8, 16, and 24, assembles P and its consistent tangent, and applies a fourth-order Duffy-product load rule. Orders six and eight provide a load-quadrature refinement check. It reports displacement L^2 , deformation-gradient, first-Piola, and minimum-determinant errors. Initialization by the identity deformation targets the intended local basin.

The distributed fixed-state test uses the cantilever geometry at a prescribed small rotation. One-, two-, and four-rank constructions share the mesh, free-space map, affine lift, state, direction, exact element Hessian, and sparse direct linear solve. The compared canonical objects are coordinates, connectivity, constraints, state, residual, CSR matrix, tangent action, and correction.

For subsequent nonlinear solves, the fixed Riesz map is the exact Hessian of (27) at $y(X) = X$ on the two-end constrained free space. Its numerical inertia record and inverse-solve true residual are part of the stopping record.

5.4 Branch-structured endpoint potential

The branch benchmark is not an incremental constitutive model. It sets previous history variables to zero and defines a scalar endpoint potential from the total engineering strain

$$\varepsilon := (\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, \gamma_{xy}, \gamma_{yz}, \gamma_{xz})^\top. \quad (30)$$

Engineering shear is converted to tensor shear before the ordered principal strains $\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$ are computed. A 10^{-15} $\text{diag}(0, 1, 2)$ perturbation supplies a deterministic tie convention at repeated values; it is not a material parameter.

For each material row, Young’s modulus $E > 0$ and Poisson ratio $-1 < \nu < 1/2$ define

$$\mu := \frac{E}{2(1 + \nu)}, \quad K := \frac{E}{3(1 - 2\nu)}, \quad \lambda_L := K - \frac{2\mu}{3} = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}. \quad (31)$$

Here c_0 , E , μ , K , and λ_L are scaled by S_0 , whereas the unit weight γ is scaled by S_0/L_0 ; ν and the material angles are dimensionless. After Davis–B strength reduction, let $\bar{c}_\lambda$ and $s_\lambda := \sin \phi_\lambda$ denote the reduced cohesion and friction parameters. The trial yield quantity is

$$f_{\text{tr}} := 2\mu((1 + s_\lambda)\varepsilon_1 - (1 - s_\lambda)\varepsilon_3) + 2\lambda_L s_\lambda \text{tr } \varepsilon - \bar{c}_\lambda. \quad (32)$$

The selected scalar potential has five labels: elastic, shear, left edge, right edge, and apex. Each plastic branch subtracts the corresponding nonnegative quadratic return term from an elastic or edge-restricted quadratic energy. Appendix A gives the denominators, multipliers, energies, and elastic-first tie order used by the code.

The assembled benchmark is

$$\Pi_h^{\text{MC}}(x; \lambda_{\text{sr}}) := \sum_{e \in \mathcal{T}_h} \sum_{q=1}^{n_q} w_{eq} \Phi_{\text{MC}}(B_{eq} A_e x; \bar{c}_{\lambda,eq}, s_{\lambda,eq}, \mu_{eq}, K_{eq}, \lambda_{L,eq}) - f_h^\top x. \quad (33)$$

The heterogeneous slope mesh contains four material regions, has bounding box $[-175, 30] \times [0, 60] \times [0, 86.6025403784]$ in scaled coordinates, and uses the glued-bottom free space. The assembled route, stopping, and discretization studies use $\lambda_{\text{sr}} = 1.55$. The fixed-element derivative comparisons instead use $\lambda_{\text{sr}} = 1.50$; the two configurations are identified separately and are never pooled. Named tetrahedral rules use 1, 11, and 24 points for P_1 , P_2 , and P_4 , respectively; an enriched 125-point Duffy rule is the fixed-state reference. Because the integrand is branchwise nonpolynomial, these names specify the discrete functional rather than an exactness claim.

The material-point matrix contains one strict interior state for every label, two-sided probes of all adjacent interfaces, rotated copies, and repeated- principal-value cases. A normalized branch margin records distance from the selected switching predicate. Ordinary Hessian claims are restricted to strict interiors. The finite-element tests add fixed P_1 , P_2 , and P_4 elements, a constructed P_2 state containing all five labels, and canonical tangent-action comparisons between the three derivative routes.

This block tests branch programming, contractions, quadrature sensitivity, and distributed derivative placement. It does not establish generalized differentiability at switches, semismooth convergence, a path-consistent plastic history, or physical slope-stability validation [23–25].

5.5 Admission sequence

Every numerical block follows the same order:

1. analytic, finite-difference, or manufactured verification;
2. canonical state, residual, matrix, and action comparison;
3. where a nonlinear endpoint is used, Riesz-scaled stopping and terminal recomputation.

Failure at one stage excludes every statement that depends on it. The reported fixed-state problems pass the first two stages on their declared scopes, and Section 7.2 states the exact endpoint and same-discretization gates for the local stopping study. None of these checks is used as performance evidence.

6 Verification

We verify the discrete formulation in stages. Independent manufactured and analytic solutions test the energies and spatial discretizations, fixed-state comparisons test the derivative routes, and canonical MPI comparisons test the distributed algebraic construction. None of these experiments is used as timing evidence.

6.1 Comparison metrics and admission gates

For two scalars, vectors, or matrices of the same physical type, we define the relative defect and the matrix-symmetry defect by

$$\delta(a, b) := \begin{cases} 0, & \|a\| = \|b\| = 0, \\ \frac{\|a - b\|}{\max(\|a\|, \|b\|)}, & \text{otherwise,} \end{cases} \quad \delta_{\text{sym}}(A) := \begin{cases} 0, & \|A\|_{\text{F}} = 0, \\ \frac{\|A - A^\top\|_{\text{F}}}{\|A\|_{\text{F}}}, & \text{otherwise.} \end{cases} \quad (34)$$

Here $\|\cdot\|$ denotes absolute value for scalars, the Euclidean norm for vectors, and the Frobenius norm for matrices. We also report absolute defects whenever either object vanishes. The canonical MPI comparison uses the nondimensional floor $\max(1, \|a\|, \|b\|)$ in its stored relative discrepancies; this convention prevents division by a nearly zero reference and is not reused as a physical endpoint norm.

The p -Laplace, Ginzburg–Landau, and nonaffine hyperelastic Newton solves use problem-specific relative weak-residual tolerances 10^{-8} , 10^{-9} , and 10^{-10} , respectively. In each case, the reported ratio is $\|R_{h,\text{free}}^{(k)}\|_2 / \|R_{h,\text{free}}^{(0)}\|_2$. We approximate the spatial errors in the L^2 norm and the H^1 seminorm by quadrature. The mechanics problem additionally uses the L^2 norm of the first-Piola stress error.

For a unit direction d , the centered Hessian-action check is

$$D_h g(x; d) := \frac{g(x + hd) - g(x - hd)}{2h}, \quad \delta(D_h g(x; d), H(x)d). \quad (35)$$

The smooth-energy, fixed-element endpoint-surrogate, and material-point checks use $h = 3 \times 10^{-5}$, 10^{-7} , and 10^{-6} , respectively. Their prescribed action tolerance is 10^{-7} ; analytic route defects must not exceed 10^{-8} , and smooth-branch symmetry defects must not exceed 10^{-10} . Complete step sweeps are retained rather than fitting their roundoff tails.

For branch-interior tests, each predicate is divided by a quantity with the same units. Yield predicates use the maximum of the cohesion, trial value, and elastic stress terms. Multiplier and interface predicates use the maximum of the principal strains, return multipliers, return tests, and $\bar{c}_\lambda / \max(|\mu|, |K|, |\lambda_L|)$. The active margin is the smallest signed normalized inequality required by the selected first-match branch; strict interiors require a margin of at least 10^{-3} .

6.2 Manufactured scalar problems

The p -Laplace problem reaches the relative weak-residual gate on all four meshes in three damped Newton steps. The successive L^2 orders are 1.986, 1.996, and 1.999, while the H^1 -seminorm orders are 0.990, 0.998, and 0.999. The finest-mesh errors are 2.76×10^{-5} and 5.45×10^{-3} , respectively. The minimum discrete gradient norm remains above 1.19, so this test avoids the zero-gradient degeneracy of the $p = 3$ Hessian.

The controlled Ginzburg–Landau branch converges in four Newton steps on every mesh. Its finest-pair orders are 2.001 in L^2 and 1.000 in the H^1 seminorm, and all nodal values lie between 0.800 and 0.902. The observed rates are consistent with the expected P_1 behavior on this selected smooth branch. They do not independently verify the quadrature rule or establish basin robustness for the zero-source double-well problem.

6.3 Hyperelastic verification

For the affine patch, the relative energy, residual, and Hessian discrepancies are 4.58×10^{-15} , 2.67×10^{-15} , and 4.38×10^{-16} , respectively. The relative boundary-traction imbalance is 1.75×10^{-16} . The net internal-force norm is 1.92×10^{-16} , and the largest rigid-translation Hessian-action norm is 1.19×10^{-15} in the nondimensional force units of the unit-cube test. A superposed rotation produces zero stored-energy defect and a 7.94×10^{-17} first-Piola covariance error.

The nonaffine manufactured problem uses 4, 8, 16, and 24 subdivisions per axis, and all four meshes converge in four damped Newton steps. The last-pair orders are 1.887 for displacement in L^2 , 1.006 for the deformation gradient, and 0.983 for first-Piola stress. The minimum computed determinant is 0.844, and the maximum tangent-symmetry defect is 9.99×10^{-17} . Across all mesh levels and the three response norms, the largest order-4-versus-order-6 solution difference is 8.29×10^{-6} times the corresponding finite-element error. The order-8 calculation separately confirms stabilization of the assembled load vector. These results support consistency on the tested orientation-preserving trajectory, but not a general nonlinear convergence rate.

6.4 Derivative and branch checks

The smooth fixed-element block contains linear, nonlinear, indefinite, and admissible finite-strain energies. Across its five cases, the maximum analytic gradient and Hessian discrepancies are 1.29×10^{-15} and 2.23×10^{-16} . The P_1 , P_2 , and P_4 endpoint-surrogate blocks each contain five prescribed element states and pass their exact CSR-structure gates. Across these blocks, the maximum residual and Hessian relative errors are 2.35×10^{-16} and 2.34×10^{-16} , respectively. For P_4 alone, the corresponding residual, Hessian, and symmetry maxima are 2.26×10^{-16} , 2.34×10^{-16} , and 1.24×10^{-16} .

The material-point matrix contains five strict branch interiors, 15 two-sided interface pairs, 15 rotated interiors, and seven repeated-principal-value cases. All finiteness, symmetry, transcription, rotation, and directional gates pass. Across the five strict interiors, the largest centered Hessian-action error at the prescribed step is 1.54×10^{-9} , and the maximum Hessian-symmetry defect is 8.68×10^{-17} . The two-sided probes exhibit order-one changes in the selected Hessian. They illustrate why the smooth-branch assumptions of Proposition 1 exclude branch interfaces; the finite probes do not prove a general impossibility result there.

6.5 Canonical distributed equivalence

The one-rank hyperelastic reference and the two- and four-rank constructions agree exactly in topology, maps, state, direction, and CSR structure. The maximum nondimensional energy, residual, matrix, tangent-action, and linear-correction discrepancies are 8.67×10^{-19} , zero, zero, 3.35×10^{-16} , and 2.33×10^{-16} , respectively. This verifies rank-local element-Hessian assembly at one prescribed state; it does not test a nonlinear endpoint or parallel efficiency.

Table 1 gives an independent check of distributed colored sparse recovery for P_1 and P_2 , elastic and mixed states, and one, two, and four ranks. All 12 blocks, comprising 36 route executions, pass the exact state, common-residual, branch, and prescribed tangent-action checks. The four one-rank blocks additionally pass exact CSR pattern and value comparisons; no direct distributed CSR comparison is claimed for two or four ranks. Before presentation rounding, the largest nonzero relative discrepancy over the checked tangent actions is 2.64×10^{-16} . The result establishes fixed-state correctness over this finite test matrix; it provides neither a scaling result nor an endpoint comparison.

6.6 Scope of the endpoint-surrogate checks

The endpoint-surrogate experiments verify the scalar branch program and its derivatives, not an independent constitutive model. The NumPy material-point code independently implements rotations, finite differences, and interface

Table 1: Fixed-state verification of distributed colored sparse recovery. State vectors agree exactly in all 12 blocks. The common element-AD residual is retained as a state and assembly consistency check. The four one-rank blocks additionally compare complete CSR patterns and values; a dash denotes that no direct distributed CSR comparison was performed. The tabulated route defects compare colored recovery with element AD at the same state and rank; constitutive-route and cross-rank discrepancies are summarized in the text. Each tabulated numerical comparison is accepted when its relative error is at most 1×10^{-8} or its absolute error is at most 1×10^{-10} . Timings are not included.

Element	State	Ranks	Common residual rel. defect	Colored/element action rel. defect	Colored/element CSR-value rel. defect
$P_1(L_1)$	Elastic	1	0	0	0
$P_1(L_1)$	Elastic	2	0	0	–
$P_1(L_1)$	Elastic	4	0	0	–
$P_1(L_1)$	Mixed	1	0	0	0
$P_1(L_1)$	Mixed	2	0	0	–
$P_1(L_1)$	Mixed	4	0	0	–
$P_2(L_1)$	Elastic	1	0	0	0
$P_2(L_1)$	Elastic	2	0	0	–
$P_2(L_1)$	Elastic	4	0	0	–
$P_2(L_1)$	Mixed	1	0	0	0
$P_2(L_1)$	Mixed	2	0	0	–
$P_2(L_1)$	Mixed	4	0	0	–

construction, but it transcribes the same endpoint energy. We therefore treat the result as internal verification. Physical validation would require a matched incremental return map and a recognized boundary-value benchmark with common history, constraints, and material assumptions.

The same limitation applies at repeated principal values. The deterministic tie convention prevents nonfinite automatic-differentiation values and selects a reproducible branch. It does not make the ordinary Hessian unique at an exact eigenvalue degeneracy.

7 Numerical Results

We now report numerical results for quadrature sensitivity, nonlinear stopping, controlled globalization, and fixed-state tangent-action agreement. These experiments clarify how the verified derivatives behave under the numerical choices used by the solver. The available local studies support correctness and bounded outcome statements, but not timing, scaling, or route-selection claims.

7.1 Fixed-state quadrature sensitivity

We evaluate one prescribed coefficient vector and one deterministic unit direction for each polynomial degree. The degree-associated tetrahedral rules use 1, 11, and 24 points for P_1 , P_2 , and P_4 , respectively; every result is compared with the same 125-point Duffy rule. Let $\Pi_q = E_q^{\text{int}} - W_q^{\text{ext}}$ and define

$$s_{\Pi,125} := \max(|E_{125}^{\text{int}}|, |W_{125}^{\text{ext}}|, |\Pi_{125}|, \text{tiny}_{64}), \quad \delta_{\Pi}(q) := \frac{|\Pi_q - \Pi_{125}|}{s_{\Pi,125}}, \quad (36)$$

where tiny_{64} is the smallest positive normal binary64 number. We report, in order, δ_{Π} , the relative free-residual defect from (34), the corresponding relative free Hessian-action defect, and the L^1 difference of the absolute-quadrature-weight branch fractions.

For P_1 , these four discrepancies are 3.51×10^{-16} , 2.79×10^{-16} , 3.60×10^{-16} , and 7.11×10^{-17} . The affine-element integrands therefore agree to roundoff. For P_2 , the corresponding values are 7.11×10^{-5} , 3.43×10^{-3} , 3.24×10^{-2} , and 1.21×10^{-2} . For P_4 , they increase to 6.28×10^{-4} , 2.68×10^{-2} , 6.91×10^{-2} , and 8.51×10^{-3} .

The energy discrepancy alone thus understates the residual and tangent sensitivity for both higher-order spaces. Because the states are prescribed, these comparisons do not measure endpoint stationarity or select a quadrature policy. The aggregate branch-fraction differences also do not locate a branch switch or establish differentiability there. A convergence conclusion requires separately converged nonlinear endpoints under each rule.

7.2 Mesh-aware stopping results

Table 2 summarizes the completed deterministic local stopping matrix. All 45 executions have checksum-identified outputs and independent numerical rechecks. Row-level admission applies a family-specific gate: Ginzburg–Landau and nonlinear rows require successful termination, setup-only hyperelastic metric checks require the positive-inertia and

Table 2: Deterministic local stopping-calibration outcomes. An accepted comparison satisfies every family-specific same-discretization gate in Appendix A.3.

Problem family	Executions	Admitted records	Accepted comparisons
Ginzburg–Landau endpoints	8/8	8/8	5/8
Hyperelasticity metric checks	6/6	6/6	6/6
Hyperelasticity nonlinear endpoints	8/8	8/8	6/8
Mohr–Coulomb surrogate, fixed state	15/15	13/15	8/15
Mohr–Coulomb surrogate, nonlinear endpoint	8/8	8/8	3/8

The reported subset contains 45 of the 52 computations in the frozen plan. Four degree-four nonlinear rows and three MPI-consistency rows remain cluster-deferred; hence the complete stopping protocol has not passed.

Table 3: Terminal outcomes for prescribed initial states in the local globalization comparison.

Problem	Method	Completed	Failed
Ginzburg–Landau, L_5 , 2 ranks	Newton–Armijo	0/15	15/15
Ginzburg–Landau, L_5 , 2 ranks	Reduced trust–Armijo	15/15	0/15
Hyperelasticity, L_2 , step 1, 2 ranks	Newton–Armijo	15/15	0/15
Hyperelasticity, L_2 , step 1, 2 ranks	Reduced trust–Armijo	15/15	0/15

Each count combines three prescribed initial states and five executions per method and initial state. These initial states form a deterministic sensitivity study, not a random sample; no population-level robustness conclusion is drawn. Timings are omitted, and the results do not establish a performance ordering. Methods are compared pairwise only when both terminate at the same independently identified endpoint.

Riesz-solve certificate, and fixed-state rows require a passed linear solve. Every record must contain its finite required diagnostics and prescribed provenance. For Ginzburg–Landau, the stored residual ratio is independently checked against its target; Riesz/Krylov rows additionally require an explicitly formed true residual within the family gate. The row-level gate admits 43 records: all Ginzburg–Landau and hyperelasticity records and all endpoint-surrogate nonlinear endpoints pass, while 13 of the 15 endpoint-surrogate fixed-state solves pass.

Within each of 11 mesh/discretization groups, the tightest computed tolerance is the reference. A candidate comparison is accepted only when all family-specific state, energy, observable, branch, and true-residual predicates in Appendix A.3 hold. Of 45 comparisons, 28 are accepted, 15 are rejected, and two are censored because their fixed-state linear endpoints fail the independent true-residual gate. The 28 accepted rows include the 11 reference self-comparisons, leaving 17 accepted nonreference candidates.

The selected largest accepted tolerances are 10^{-6} and 10^{-4} for the two Ginzburg–Landau meshes; 10^{-8} for both hyperelastic metric checks; 10^{-4} for both hyperelastic nonlinear levels; 10^{-6} , 10^{-6} , and 10^{-8} for the degree-one, degree-two, and degree-four fixed-state surrogate solves; and 10^{-6} and 10^{-7} for the degree-one and degree-two nonlinear surrogate solves. These are deterministic same-discretization calibrations, not mesh-convergence evidence or a universal stopping policy.

The reported subset contains 45 of the 52 computations in the frozen stopping plan. Four degree-four nonlinear rows and three MPI-consistency rows remain cluster-deferred. Hence the local calibration is admissible, but the complete stopping protocol has not passed; closing those rows is necessary but not sufficient without the separate discretization gate and final adjudication.

7.3 Controlled globalization outcomes

The controlled design uses three prescribed starts and five separate executions for each method and problem, giving the 60 two-rank outcomes in Table 3. For the Ginzburg–Landau problem on L_5 , Newton–Armijo fails in all 15 executions, whereas reduced trust–Armijo completes all 15. For the first hyperelastic load step on L_2 , both methods complete all 15 executions.

A paired comparison is admitted only if both methods terminate successfully and produce exactly the same canonical endpoint. The gate compares dtype-, shape-, and byte-aware SHA-256 digests of both the exported global state and the canonically ordered solver vector; it uses no positive numerical tolerance. All 30 common-start checks pass. The 15 Ginzburg–Landau pairs are ineligible for endpoint comparison because Newton–Armijo fails at its first iteration after 40 Armijo evaluations in every execution, whereas reduced trust–Armijo completes. All 15 first-load hyperelastic pairs reach endpoint comparison, but none passes exact digest equality. Consequently, the experiment supports only the displayed completion and failure counts. It does not support a paired method ranking, a robustness probability, or a timing comparison. Because this is an exact reproducibility gate rather than a prespecified state–energy–observable distance, its failure alone does not establish distinct physical endpoint classes.

7.4 Fixed-state tangent-action agreement

The checksum-verified workstation study compares element automatic differentiation, constitutive automatic differentiation, and colored sparse recovery for $P_1(L_1)$ and $P_2(L_1)$ at elastic and mixed states. The 12 balanced blocks contain 36 sequential route executions, and each of the six route orders occurs twice. All blocks complete, and the declared output inventory passes its independent checksum verification.

Across four deterministic probes per route and state, the largest relative tangent-action discrepancy is 2.34×10^{-16} . The saved states and the common element-AD residuals agree exactly at stored precision. This finite study therefore corroborates four-probe tangent-action agreement on the tested serial states. No route timing is reported, and the finite action checks imply no performance ordering.

Together, these experiments quantify higher-order quadrature sensitivity, establish the local extent of the stopping checks, record bounded globalization outcomes, and verify serial four-probe tangent-action agreement. Their explicit scope excludes nonlinear quadrature convergence, a general stopping or globalization policy, parallel scaling, and performance-based route selection.

8 Discussion

The analysis distinguishes a mathematical identity from the conditions needed to realize it in distributed software. On a fixed smooth branch, element and constitutive differentiation are two applications of the chain rule to the same finite-dimensional potential. Colored recovery adds genuinely algebraic requirements: a complete pattern, noninterfering same-color columns, consistent overlap values, and unique ownership of every recovered row. Failure of any one condition can preserve the scalar potential while corrupting the sparse operator.

The finite-element experiments support both conditional statements on their declared scopes. Element and constitutive routes agree across 15 prescribed P_1 , P_2 , and P_4 states, with maximum residual and Hessian relative discrepancies of 2.35×10^{-16} and 2.34×10^{-16} . The separate colored-recovery study covers $P_1(L_1)$ and $P_2(L_1)$, elastic and mixed states, and one, two, and four ranks. Direct CSR comparisons pass in the four one-rank blocks, and the maximum cross-rank action discrepancy is 2.64×10^{-16} . These are strong regression tests for the stated discrete objects, but they are not proofs for arbitrary meshes, constraint maps, partitions, or future implementation changes.

The quadrature study shows why the discrete functional must be part of every derivative claim. Relative to the 125-point reference, the degree-associated $P_2(L_1)$ rule changes the potential scale by only 7.11×10^{-5} , while the free residual and Hessian action change by 3.43×10^{-3} and 3.24×10^{-2} . For $P_4(L_1)$ the corresponding values are 6.28×10^{-4} , 2.68×10^{-2} , and 6.91×10^{-2} . No sampled P_2 point lies inside the declared 10^{-8} switch band. The discrepancies therefore quantify fixed-state quadrature sensitivity on the selected branches; they neither establish a degree trend nor replace separately converged nonlinear endpoints.

The Mohr–Coulomb endpoint tests have an equally specific boundary. All five selected labels are exercised at strict interior points, and the two-sided interface, rotation, and repeated-spectrum probes remain finite under the declared tie convention. Ordinary gradients and Hessians are used only within branch interiors. No generalized derivative, semismooth convergence result, incremental history validation, or physical slope-stability validation follows from this surrogate.

The nonlinear studies illustrate why solver outcomes and method comparisons must be reported separately. All 45 local stopping cases completed; 43 records and 28 same-discretization comparisons pass their explicitly stated gates. The seven cluster-deferred rows prevent a complete 52-row stopping-protocol pass. These deterministic calibrations do not establish mesh convergence, a universal stopping tolerance, or rank independence. In the 60-execution globalization study, Newton–Armijo fails on all 15 Ginzburg–Landau executions, whereas the reduced-trust method completes all 15; both methods complete all 15 first-load hyperelastic executions. The Ginzburg–Landau pairs never become eligible for endpoint comparison, and none of the 15 hyperelastic pairs passes the exact endpoint-identity condition. These counts therefore support neither a method ordering nor a robustness probability. Failure of the zero-tolerance hash gate also does not establish distinct physical endpoint classes.

The checksum-verified workstation study verifies four-probe tangent-action agreement rather than performance. It comprises 12 blocks and 36 sequential route executions and limits the maximum action discrepancy to 2.34×10^{-16} . No route timing enters the analysis, so the data support no speedup, crossover, or route-ordering statement.

The independent manufactured problems verify consistency rather than physical validity of the production benchmarks. The scalar solutions are smooth, the Ginzburg–Landau verifier stays on a positive-curvature branch, and the hyperelastic tests retain positive determinant. Except for the strictly convex p -Laplace problem, small first-order residuals do not certify a local or global minimizer. The main conclusion is therefore deliberately limited: derivative placement can be changed without changing the tested discrete operator when the mathematical and distributed contracts are both satisfied.

9 Conclusions

We established a conditional equivalence framework for three placements of automatic differentiation in nonlinear finite elements. Element and constitutive differentiation agree when they act on the same quadrature-defined potential, affine

constraint map, strain convention, and smooth active branch. Colored sparse recovery produces the same operator only when the pattern, interference coloring, overlap state, and global row ownership are also consistent.

The checksum-verified numerical evidence supports these statements on the tested problems. Element and constitutive routes agree to binary64 precision across 15 prescribed linear and high-order states. Distributed colored recovery passes the prescribed tangent-action checks for $P_1(L_1)$ and $P_2(L_1)$ elastic and mixed states on one, two, and four ranks; direct CSR comparisons pass at one rank. Independent manufactured and analytic checks recover the expected scalar finite-element behavior and verify smooth hyperelastic stresses and tangents. The branch program passes its interior, interface, rotation, and repeated- spectrum checks within the explicit exclusion of ordinary Hessians at switches.

The same evidence also identifies the limits of the conclusions. Fixed-state quadrature changes residuals and Hessian actions much more than the potential scale for P_2 and P_4 . The completed 45-row local stopping matrix calibrates only deterministic same-discretization accuracy; seven cluster-deferred rows prevent a complete stopping-protocol pass. In the globalization study, the Ginzburg–Landau pairs do not reach endpoint comparison and none of the hyperelastic pairs passes exact endpoint identity. The workstation route study passes its state, common-residual, four-probe action, and provenance checks, but no route timing enters the analysis. Consequently, we claim no fastest derivative route, crossover predictor, nonlinear robustness probability, or parallel scaling law.

A Branch and Solver Details

A.1 Complete endpoint-branch definition

This subsection records the exact scalar branch program used in Section 5.4. Let c_0 , ϕ , and ψ denote the input cohesion, friction angle, and dilation angle. For strength-reduction factor $\lambda_{\text{sr}} > 0$, the Davis–B map is defined as in the stated slope-stability reduction context [26, 32, 33]:

$$\begin{aligned} c_1 &:= \frac{c_0}{\lambda_{\text{sr}}}, & \phi_1 &:= \arctan\left(\frac{\tan \phi}{\lambda_{\text{sr}}}\right), & \psi_1 &:= \arctan\left(\frac{\tan \psi}{\lambda_{\text{sr}}}\right), \\ \beta &:= \frac{\cos \phi_1 \cos \psi_1}{1 - \sin \phi_1 \sin \psi_1}, & c_\lambda &:= \beta c_1, & \phi_\lambda &:= \arctan(\beta \tan \phi_1), \end{aligned} \quad (37)$$

with

$$\bar{c}_\lambda := 2c_\lambda \cos \phi_\lambda, \quad s_\lambda := \sin \phi_\lambda. \quad (38)$$

The tested material table has $\psi = 0$, but the implementation retains the complete map (37). The mathematical domain used below has finite ordered principal strains, $K > 0$, $\mu > 0$, $-1 < s_\lambda < 1$, and $s_\lambda \neq 0$ whenever the apex row is reachable. On this domain the displayed denominators are nonzero. Defensive tiny-denominator substitutions in the program are failure containment outside this model; an admitted row must instead pass the stored denominator-margin gate.

Let $I_1 := \text{tr } \varepsilon$ and

$$Q(\varepsilon) := \varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2 + \frac{1}{2}(\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{xz}^2). \quad (39)$$

The elastic energy is

$$\Phi_{\text{el}} := \frac{1}{2}\lambda_{\text{L}}I_1^2 + \mu Q(\varepsilon). \quad (40)$$

In addition to f_{tr} from (32), the return tests are

$$\begin{aligned} r_{\text{sl}} &:= \frac{\varepsilon_1 - \varepsilon_2}{1 + s_\lambda}, & r_{\text{sr}} &:= \frac{\varepsilon_2 - \varepsilon_3}{1 - s_\lambda}, \\ r_{\text{la}} &:= \frac{\varepsilon_1 + \varepsilon_2 - 2\varepsilon_3}{3 - s_\lambda}, & r_{\text{ra}} &:= \frac{2\varepsilon_1 - \varepsilon_2 - \varepsilon_3}{3 + s_\lambda}. \end{aligned} \quad (41)$$

Their strictly positive denominators are

$$\begin{aligned} D_{\text{s}} &:= 4\lambda_{\text{L}}s_\lambda^2 + 4\mu(1 + s_\lambda^2), \\ D_1 &:= 4\lambda_{\text{L}}s_\lambda^2 + \mu(1 + s_\lambda)^2 + 2\mu(1 - s_\lambda)^2, \\ D_{\text{r}} &:= 4\lambda_{\text{L}}s_\lambda^2 + 2\mu(1 + s_\lambda)^2 + \mu(1 - s_\lambda)^2, \\ D_{\text{a}} &:= 4Ks_\lambda^2. \end{aligned} \quad (42)$$

For the tested parameters $s_\lambda > 0$; the verifier records normalized denominator margins.

The four plastic multipliers are

$$\begin{aligned}
\Lambda_s &:= \frac{f_{\text{tr}}}{D_s}, \\
\Lambda_l &:= \frac{\mu((1+s_\lambda)(\varepsilon_1+\varepsilon_2) - 2(1-s_\lambda)\varepsilon_3) + 2\lambda_L s_\lambda I_1 - \bar{c}_\lambda}{D_l}, \\
\Lambda_r &:= \frac{\mu(2(1+s_\lambda)\varepsilon_1 - (1-s_\lambda)(\varepsilon_2+\varepsilon_3)) + 2\lambda_L s_\lambda I_1 - \bar{c}_\lambda}{D_r}, \\
\Lambda_a &:= \frac{2K s_\lambda I_1 - \bar{c}_\lambda}{D_a}.
\end{aligned} \tag{43}$$

The associated energies are

$$\begin{aligned}
\Phi_s &:= \Phi_{\text{el}} - \frac{1}{2} D_s \Lambda_s^2, \\
\Phi_l &:= \frac{1}{2} \lambda_L I_1^2 + \mu \left(\varepsilon_3^2 + \frac{1}{2} (\varepsilon_1 + \varepsilon_2)^2 \right) - \frac{1}{2} D_l \Lambda_l^2, \\
\Phi_r &:= \frac{1}{2} \lambda_L I_1^2 + \mu \left(\varepsilon_1^2 + \frac{1}{2} (\varepsilon_2 + \varepsilon_3)^2 \right) - \frac{1}{2} D_r \Lambda_r^2, \\
\Phi_a &:= \frac{1}{2} K I_1^2 - \frac{1}{2} D_a \Lambda_a^2.
\end{aligned} \tag{44}$$

The elastic-first selection order is

$$\Phi_{\text{MC}}(\varepsilon) := \begin{cases} \Phi_{\text{el}}, & f_{\text{tr}} \leq 0, \\ \Phi_s, & f_{\text{tr}} > 0 \text{ and } \Lambda_s \leq \min(r_{\text{sl}}, r_{\text{sr}}), \\ \Phi_l, & f_{\text{tr}} > 0 \text{ and } r_{\text{sl}} < r_{\text{sr}} \text{ and } r_{\text{sl}} \leq \Lambda_l \leq r_{\text{la}}, \\ \Phi_r, & f_{\text{tr}} > 0 \text{ and } r_{\text{sr}} < r_{\text{sl}} \text{ and } r_{\text{sr}} \leq \Lambda_r \leq r_{\text{ra}}, \\ \Phi_a, & f_{\text{tr}} > 0 \text{ and none of the three tests above holds,} \end{cases} \tag{45}$$

The cases are evaluated from top to bottom and the first true row is returned; mutual exclusivity of the un-ordered plastic predicates is therefore not assumed. The predicates are part of the discrete surrogate. Differentiation treats the selected row of (45) as fixed.

A.2 Interface and repeated-spectrum probes

For every adjacent branch pair, the material-point verifier constructs states on both sides of the switching predicate. Each state records the normalized active-branch margin, constitutive denominator, principal-value gap, finite stress and tangent, tangent symmetry, and centered finite-difference action. The two one-sided selected Hessians are compared descriptively. Their difference does not define a generalized derivative.

Repeated principal values are evaluated under the deterministic diagonal perturbation stated in Section 5.4. The checks require finite energy, stress, and tangent, but they do not claim rotational invariance of the arbitrary tie-selected tangent at exact degeneracy. Separate rotations of strict interior states test energy invariance and stress/tangent covariance.

A.3 Stopping-study scope

The local stopping matrix contains 45 checksum-identified cases. Its parameter grid is as follows: Ginzburg–Landau relative residual targets $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$ on L_5 and L_6 ; hyperelastic Riesz-solve relative tolerances $10^{-8}, 10^{-10}, 10^{-12}$ on L_1 and L_2 ; hyperelastic nonlinear relative residual targets $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$ on the same two levels; fixed-state endpoint-surrogate Krylov tolerances $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}$ for degrees one, two, and four; and endpoint-surrogate nonlinear relative residual targets $10^{-2}, 10^{-4}, 10^{-6}, 10^{-7}$ for degrees one and two. The reported 45-row local subset contains no degree-four nonlinear endpoint and supports no inference about rank-count consistency. The four degree-four nonlinear rows and three MPI-consistency rows in the 52-row frozen plan remain cluster-deferred.

For every mesh/discretization group, let $x_\star$ denote the result at the tightest computed tolerance and x_τ a candidate. When a scalar relative difference is required, we use

$$d_{\text{rel}}(a, b) := \frac{|a - b|}{\max(|a|, |b|, \text{tiny}_{64})}. \tag{46}$$

For Ginzburg–Landau, the same-mesh state discrepancy is

$$\eta_{L_h^2} := \frac{[\sum_i m_i (u_{\tau,i} - u_{\star,i})^2]^{1/2}}{\max([\sum_i m_i u_{\star,i}^2]^{1/2}, \text{tiny}_{64})}, \quad m_i := \sum_{T \ni i} \frac{|T|}{3}. \tag{47}$$

For mechanics, with the frozen reference-elastic Riesz operator K_{ref} , the state discrepancy is

$$\eta_K := \frac{[(x_\tau - x_\star)^\top K_{\text{ref}}(x_\tau - x_\star)]^{1/2}}{\max([x_\star^\top K_{\text{ref}} x_\star]^{1/2}, \text{tiny}_{64})}. \quad (48)$$

The fixed-state linear comparison applies the same formula to the correction vectors. The coefficient-space discrepancy is

$$\eta_2(v_\tau, v_\star) := \frac{\|v_\tau - v_\star\|_2}{\max(\|v_\star\|_2, \text{tiny}_{64})}, \quad (49)$$

where v is the displacement array for a hyperelastic endpoint and the correction vector for a fixed-state endpoint-surrogate solve. We write $\Pi_\tau := \Pi_h(x_\tau)$, $\Pi_\star := \Pi_h(x_\star)$, $\omega_\tau := f_h^\top x_\tau$, and $\omega_\star := f_h^\top x_\star$. The maximum nodal displacements are $u_{\max, \tau} := \max_i \|\mathbf{u}_{\tau, i}\|_2$ and $u_{\max, \star} := \max_i \|\mathbf{u}_{\star, i}\|_2$.

Record admission is distinct from comparison admission. Ginzburg–Landau requires one successful solve/load record, the lumped- L^2 residual metric, and a stored relative residual independently checked against its row target. Every reference-elastic norm solve uses conjugate gradients with Jacobi preconditioning, the unpreconditioned residual norm, zero absolute tolerance, and a cap of 5,000 iterations; the requested and effective settings must agree, and runtime options may not alter them. Hyperelastic Riesz-setup rows require a positive numerical-inertia certificate for K_{ref} , a positive Krylov convergence reason, and a relative true residual no larger than 10^{-6} . Hyperelastic nonlinear and nonlinear endpoint-surrogate rows require the same certificate and convergence reason with a true-residual gate of 10^{-8} . A nonlinear endpoint must also satisfy $R_k/R_0 \leq \tau_R^{\text{rel}}$. A fixed-state endpoint-surrogate row requires a positive Krylov reason, branch margin at least 10^{-8} , zero quadrature fraction in the switch band, and relative true residual no larger than $\max(10^{-12}, 20\tau_{\text{Krylov}})$.

Subject to endpoint admission, every condition in the applicable comparison row below is conjunctive:

1. Ginzburg–Landau requires $\eta_{L_h^2} \leq 10^{-5}$ and $|\Pi_\tau - \Pi_\star| \leq 10^{-10}$.
2. A hyperelastic Riesz setup check requires relative dual-norm and state-scale differences no larger than 10^{-6} and 10^{-12} , respectively, and a relative true residual no larger than $\max(10^{-8}, 20\tau_{\text{Riesz}})$.
3. A hyperelastic nonlinear endpoint requires η_2 and η_K to be no larger than 10^{-5} , and $|\Pi_\tau - \Pi_\star| \leq 10^{-8}$.
4. A fixed-state endpoint-surrogate solve requires η_2 and the reference-elastic relative correction discrepancy to be no larger than 10^{-4} , and a relative true residual no larger than $\max(10^{-10}, 20\tau_{\text{Krylov}})$.
5. A nonlinear endpoint-surrogate solve requires $\eta_K \leq 10^{-5}$, $|\Pi_\tau - \Pi_\star| \leq 10^{-6}$, $|\omega_\tau - \omega_\star| \leq 10^{-6}$, $|u_{\max, \tau} - u_{\max, \star}| \leq 10^{-8}$, and exact equality of the complete branch-count dictionaries.

The two inadmissible records are degree-four fixed-state solves. Their independent relative true residuals are $0.305 > 0.2$ at $\tau_{\text{Krylov}} = 10^{-2}$ and $2.20 \times 10^{-5} > 2.00 \times 10^{-5}$ at $\tau_{\text{Krylov}} = 10^{-6}$. Among the 15 rejected but comparable rows, the failed-predicate incidences are: Ginzburg–Landau state 2 and energy 3; hyperelastic nonlinear coefficient, Riesz-state, and energy 2 each; fixed-state surrogate coefficient and Riesz-correction 5 each; and nonlinear surrogate Riesz-state 2, energy 4, external work 5, maximum displacement 4, and branch counts 3. These incidences are nonexclusive because one row may fail several predicates. Thus the local study calibrates deterministic same-discretization accuracy only; it is neither a mesh-convergence study nor a timing, scaling, or population-robustness experiment.

A.4 Numerical configurations

The scalar manufactured meshes have 8, 16, 32, and 64 uniform subdivisions per coordinate direction. The p -Laplace solve starts from the affine extension $x + y$ and the Ginzburg–Landau solve starts from its exact Dirichlet extension; both use damped Newton iteration and the relative residual defined in Section 6.1. The nonaffine hyperelastic meshes and material constants are stated in Section 5.3. The affine patch uses one unit cube divided into six tetrahedra.

The controlled zero-source Ginzburg–Landau study uses the level-5 P_1 mesh of $[-1, 1]^2$, with 4,225 nodes, 8,192 triangles, and 3,969 free coefficients. It sets $\epsilon = 0.01$, $f = 0$, and $u = 0$ on the complete boundary. The nominal free state is

$$u_0(x, y) := \sin\left(\frac{\pi(x-1)}{2}\right) \sin\left(\frac{\pi(y-1)}{2}\right). \quad (50)$$

The two additional starts are $u_0 \pm 0.025 m_{\text{GL}}$, where

$$m_{\text{GL}}(x, y) := \sin\left(\frac{\pi(x+1)}{2}\right) \sin\left(\frac{\pi(y+1)}{2}\right) \sin(\pi(x+1)) \sin(\pi(y+1)). \quad (51)$$

The perturbation is applied only to free nodes and vanishes on the boundary. Each state is stored once in a checksum-identified NumPy archive and is shared by both methods and all five executions. The comparison uses two MPI ranks, one thread per rank, element Hessians, GMRES with hypre preconditioning, relative tolerance 10^{-3} and cap 200, and at most 100 nonlinear iterations. The three starts are deterministic sensitivity cases rather than population samples.

The distributed hyperelastic fixed-state test uses the brick $[0, 0.4] \times [-0.005, 0.005]^2$, an $80 \times 2 \times 2$ cell grid split into 1,920 tetrahedra, 729 nodes, and 2,133 free vector DOFs after both end faces are constrained. Its constants are

Table 4: Scaled material parameters for the dry heterogeneous three-dimensional slope. Saturated and unsaturated unit weights coincide in this test.

Region	c_0/S_0	ϕ (deg)	ψ (deg)	E/S_0	ν	$\gamma L_0/S_0$
Cover layer	15	30	0	10,000	0.33	19
General foundation	15	38	0	50,000	0.30	22
Weak foundation	10	35	0	50,000	0.30	21
Slope mass	18	32	0	20,000	0.33	20

$C_1 = 3.8462 \times 10^7$ and $D_1 = 8.3333 \times 10^7$ in the nondimensional stress scale of Section 5.3; run records retain the full binary64 values. The stored fixed state twists the right face by 0.15 radians and is assembled independently on one, two, and four ranks; no nonlinear trajectory is inferred from that state.

The branch-structured endpoint benchmark uses the L_1 macro mesh with 3,845 vertices and 18,419 tetrahedra. Its bounding box is $[-175, 30] \times [0, 60] \times [0, 86.6025403784]$ in the scaled coordinates of Section 5.3. Define

$$\begin{aligned}\Gamma_b &:= \partial\Omega \cap \{y = 0\}, \\ \Gamma_x^- &:= \partial\Omega \cap \{x = -175\}, \quad \Gamma_x^+ := \partial\Omega \cap \{x = 30\}, \\ \Gamma_z^- &:= \partial\Omega \cap \{z = 0\}, \quad \Gamma_z^+ := \partial\Omega \cap \{z = 86.6025403784\}.\end{aligned}\tag{52}$$

For $\mathbf{u} = (u_x, u_y, u_z)$, the homogeneous essential conditions are

$$\mathbf{u} = \mathbf{0} \text{ on } \Gamma_b, \quad u_x = 0 \text{ on } \Gamma_x^- \cup \Gamma_x^+, \quad u_z = 0 \text{ on } \Gamma_z^- \cup \Gamma_z^+.\tag{53}$$

All remaining displacement components are free. No surface-load term is applied; hence the fully free boundary has zero natural traction, and the unconstrained components of the roller planes have zero natural tangential traction. The body force is $\mathbf{b}(X) = -\gamma(X)\mathbf{e}_y$, with γ piecewise constant by material region. All prescribed displacement values vanish, so the affine lift is zero.

The P_1 , P_2 , and P_4 constrained spaces have 10,526, 79,024, and 610,964 free vector DOFs. The four material rows are given in Table 4. Here c_0 and E are scaled by S_0 , γ is scaled by S_0/L_0 , and ϕ , ψ , and ν are dimensionless. The derived elastic moduli follow (31). The archived evidence identifies the source-mesh contents by a SHA-256 digest.

The fixed-element derivative comparison uses $\lambda_{\text{sr}} = 1.50$, whereas the assembled route, stopping, and quadrature studies use $\lambda_{\text{sr}} = 1.55$. These configurations remain separate in every numerical comparison.

The controlled globalization study uses two MPI ranks and one thread per rank. Both Newton–Armijo and reduced-subspace trust with Armijo acceptance use the same stored initial state for each of the three starts and are used in five separate executions. Ginzburg–Landau uses the coefficient ℓ^2 metric, GMRES with hypre preconditioning, relative tolerance 10^{-3} and cap 200, and at most 100 nonlinear iterations.

The hyperelastic study uses the procedural level-2 P_1 grid with $160 \times 4 \times 4$ brick cells, split into 15,360 tetrahedra, on $[0, 0.4] \times [-0.005, 0.005]^2$. It has 12,075 total and 11,925 free vector DOFs, $C_1 = 3.8461538461538464 \times 10^7$, and $D_1 = 8.333333333333333 \times 10^7$. The left face retains the identity map and the right face rotates by $\pi/3$ about the x -axis. The nominal free coefficients are initialized from the identity deformation, while the prescribed rotated-boundary lift is imposed exactly. The two additional starts add $\pm 10^{-5} m_{\text{HE}}$ on the free components, where

$$m_{\text{HE}}(X) := \left(0, \sin \frac{\pi X_1}{0.4}, -\frac{1}{2} \sin \frac{\pi X_1}{0.4}\right).\tag{54}$$

Both methods use coefficient ℓ^2 stopping and GMRES with PETSc’s algebraic multigrid preconditioner, relative tolerance 10^{-1} , cap 30, and at most 100 nonlinear iterations. Initial and terminal states are identified by SHA-256 digests.

For every retained globalization row, Armijo uses initial step 1, sufficient-decrease constant 10^{-4} , contraction $1/2$, and at most 40 trials. The reduced trust model uses initial, minimum, and maximum radii 1, 10^{-8} , and 10^6 , shrink/expand factors $1/2$ and $3/2$, acceptance-update thresholds 0.05 and 0.75, and at most six radius-reduction retries after the initial candidate, for seven candidate attempts in total. Section 7.3 states why none of these rows supports a timing comparison.

The local calculations used an AMD Ryzen Threadripper PRO 7975WX processor and the Linux 7.1.2 kernel on x86-64. The software environment contained Python 3.12.10, NumPy 2.4.2, JAX/JAXlib 0.9.1, petsc4py 3.24.2, and Open MPI 5.0.10, with one thread for OpenMP and the basic linear algebra subprograms (BLAS). Every retained archive binds the source commit, inputs, configuration, command, environment variables, outputs, and referenced state artifacts by SHA-256. The public data deposit must preserve these manifests together with the run records.

Code and Data Availability

The implementation, checked-in mesh inputs, experiment protocols, and table-generation scripts are available in the development repository https://github.com/Beremi/fenics_nonlinear_energies. The clean run archives

summarized in this paper are not tracked by that repository and have not yet been deposited in a public archive. A submission release must therefore include an immutable archive of the run records, canonical states, manifests, and checksums, together with a persistent identifier. No archival digital object identifier (DOI) is claimed here.

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